41107 Marin and ocean engineering

L9: The ocean environment - Waves

Today's content

- The ocean environment: Wind, **waves**, and ice
- Waves
  - Basic concepts
  - Regular (sinusoidal) waves; covered already(!)
  - Irregular waves
  - Statistical concepts and assumptions
- **The wave energy density spectrum**
  - **Frequency domain representation**
  - **Spectral moments and statistics**
  - **Parameterised wave spectra**
- Ocean waves
  - Statistics
  - Wave measurements and predictions (forecast/hindcast)
  - Design conditions

The program

- Lecture (ocean waves)
- Introduction to Assignment B
- Work on Assignment B (today, focus on only part about waves)

Reading:

- Chapters 10-12 in the lecture notes (Nielsen, 2024)

Ocean = air and water

- Structures in the sea are exposed to loads from air and water
  - Wind
  - Waves
  - Ice
- They can all be associated with potential catastrophic consequences
- Wind and waves induce “global” loads while ice often is a “local” load
- … but due to water’s (in)compressibility severe local slamming loads may occur
- Sea currents and tides

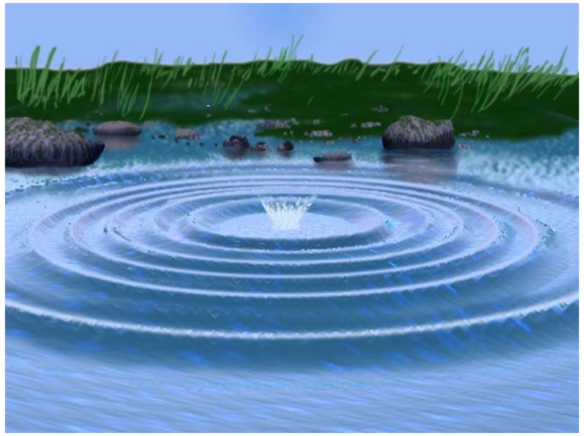
Wind may also be of *assistance…*

The forming of waves

How are waves in the oceans generated?

- The "preferred" state would be a flat sea surface (this takes minimum energy)…
- As such, waves are formed because of a disturbance that supply energy:
  1. Wind
  2. Gravitational effects (tides)
  3. Geological activity (tsunami)

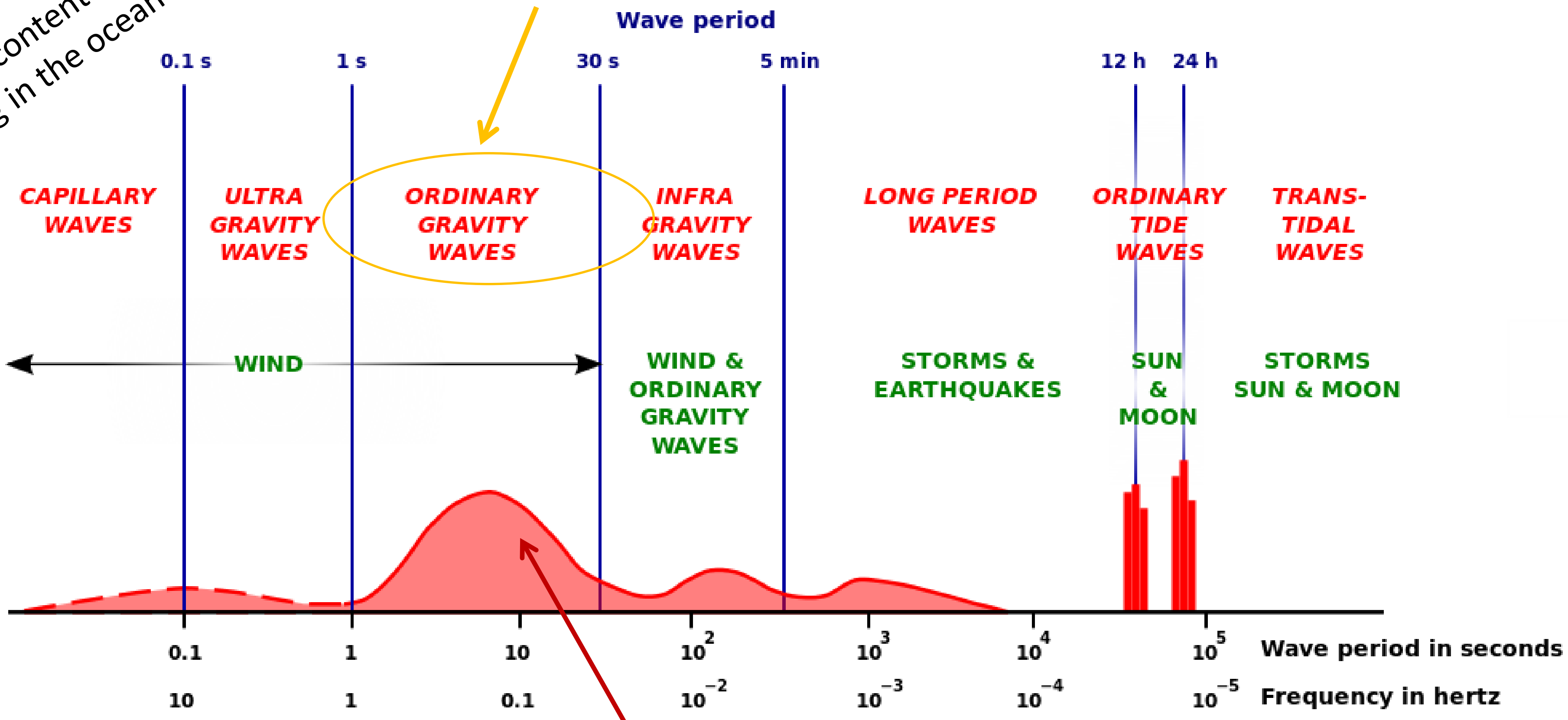
  In either case, gravity restores equilibrium ---> ***gravity waves***
- Our focus will be on the wind-generated waves



Types of waves in the ocean

Representation of energy content of different types of waves occurring in the ocean

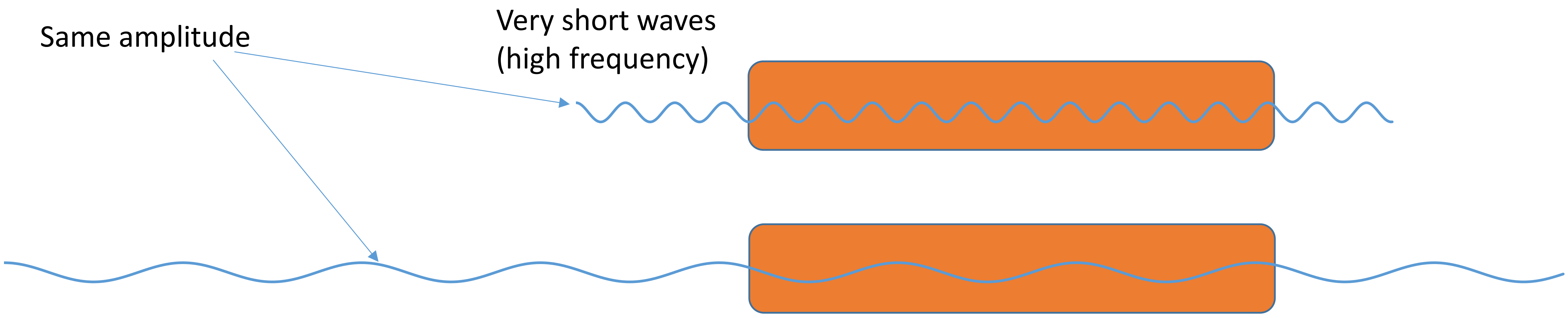
Wave frequencies of interest for ships and floating structures. Why?



These waves (i.e. their energy distribution) are the most important to floating structures.

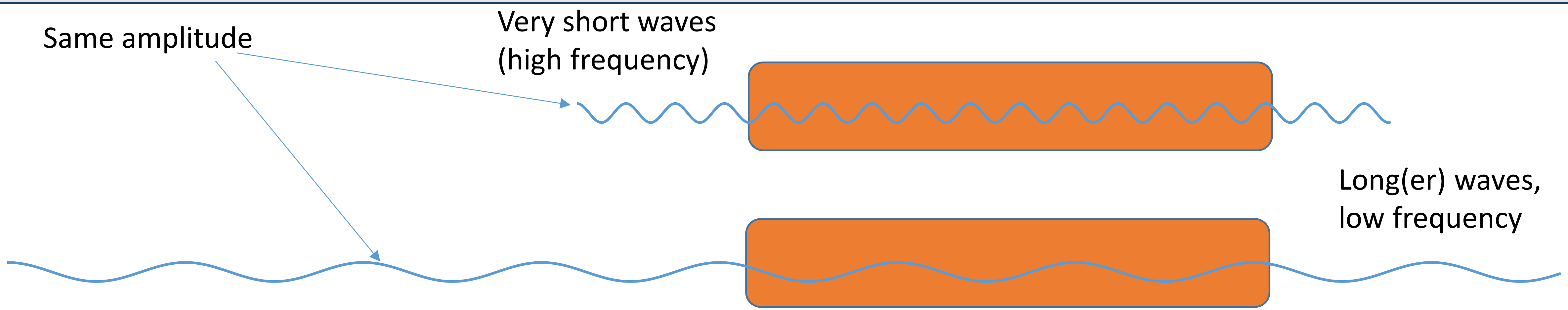
Give it a thought…

- Why is it (primarily) waves with a period 5-20 seconds which is relevant for most floating structures?
- In which situation, will the ship respond the most?



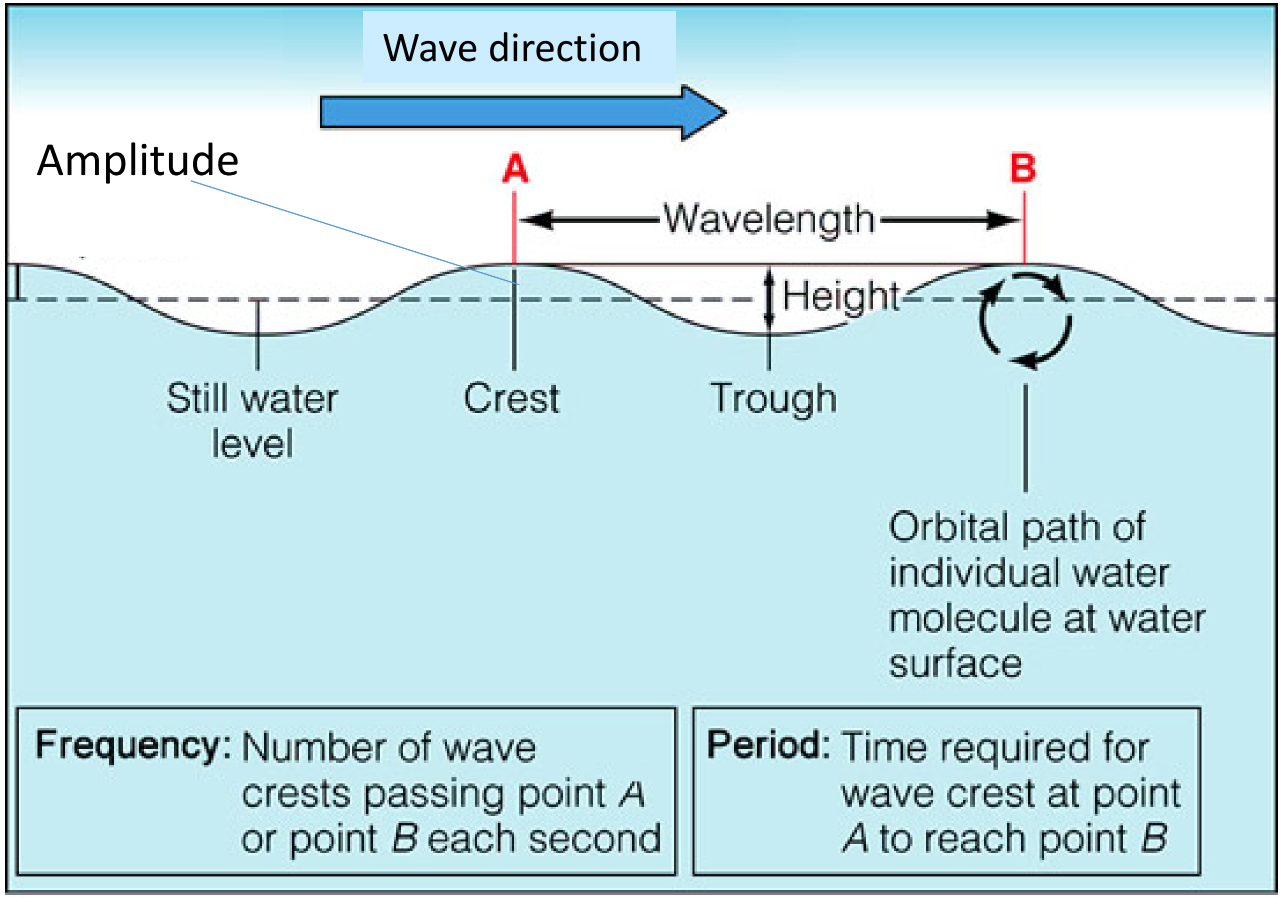
The answer

- Why is it (primarily) waves with a period 5-20 seconds which is relevant for most floating structures?



Regular ("small amplitude") waves

A conceptual illustration of a regular wave



- Waves are usually characterized by,
  - An amplitude: half the height difference between the crest (top) and the trough (bottom)
  - A period: the *time* between one wave crest and the next
  - A wave length: the *distance* between one wave crest and the next
  - A speed of propagation (celerity): how fast does the crest of the wave move

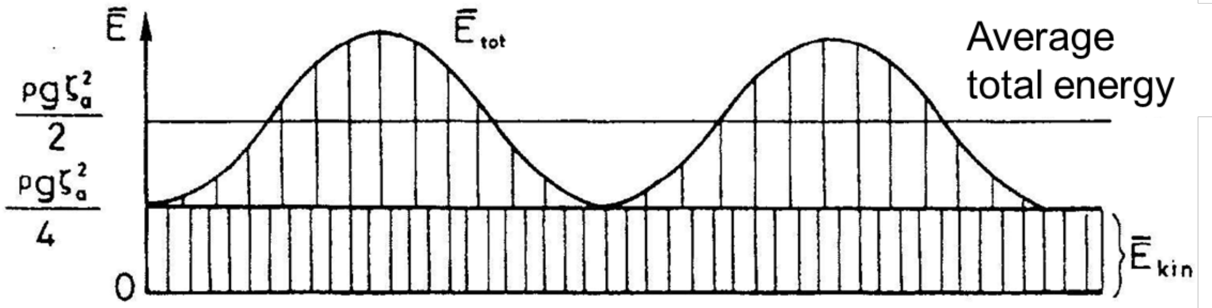
NB: Refer to lecture 3 ('basic wave theory')

Regular waves – Energy

- The *total* energy carried by regular waves consists of kinetic energy (particle motions) and potential energy (change of water level)
- It can be shown that the energy is (per unit width):

$$\bar{E}_{pot} = \frac{1}{4}\rho g\, \zeta_a^2 L \qquad \bar{E}_{kin} = \frac{1}{4}\rho g\, \zeta_a^2 L$$

- Thus, **the total energy** per unit breadth is: $\boxed{\bar{E}_{tot} = \frac{1}{2}\rho g\, \zeta_a^2 L}$

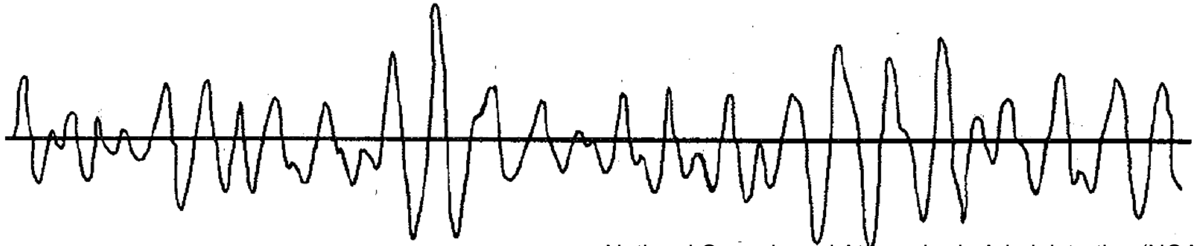


NB: Refer also to lecture 3 ('basic wave theory').

Regular waves

… but why do we have an interest in the first place?

- Clearly, *real* waves observed in the ocean are not regular but highly irregular!



National Oceanic and Atmospheric Administration (NOAA)

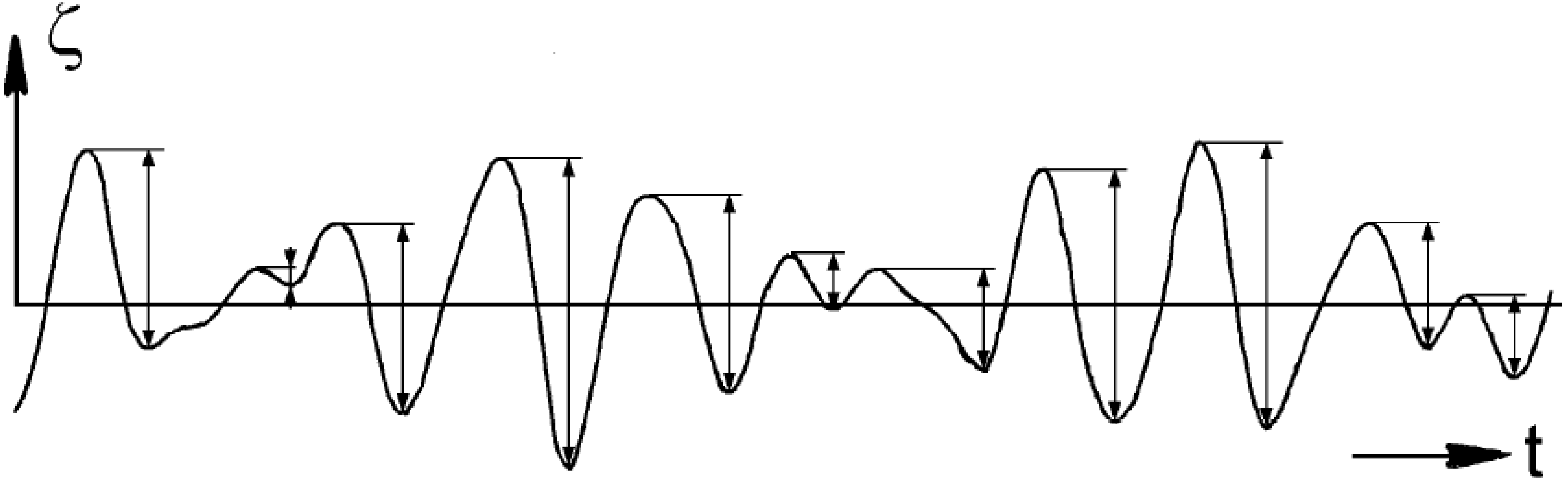
Ocean waves = sea state…

Characterisation of ocean waves by *sea state*

- Definition of **sea state** (*Wikipedia*):
- “In oceanography, a sea state is the general condition of the free surface on a large body of water — with respect to wind waves and swell — at a certain location and moment.”
- “A sea state is characterized by statistics, including the [characteristic] wave height and period, [often supplemented with a] power spectrum. The sea state varies with time, as the wind conditions or swell conditions change.”
- “The sea state can either be assessed by an experienced observer, like a trained mariner, or through instruments like weather buoys, wave radar or remote sensing satellites.”

Characterisation of ocean waves by *sea state*

- ***Wave height and period*** ... But what height and period to use? They are all different!



Significant wave height: H_s (or $H_{1/3}$)

Characterisation by *sea state*

- **Example**: Individual heights and corresponding periods (sorted)

rank	i	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
H	(m)	5.5	4.8	4.2	3.9	3.8	3.4	2.9	2.8	2.7	2.3	2.2	1.9	1.8	1.1	0.23
T	(s)	12.5	13.0	12.0	11.2	15.2	8.5	11.9	11.0	9.3	10.1	7.2	5.6	6.3	4.0	0.9
Wave No.		7	12	15	3	5	4	2	11	6	1	10	8	13	14	9

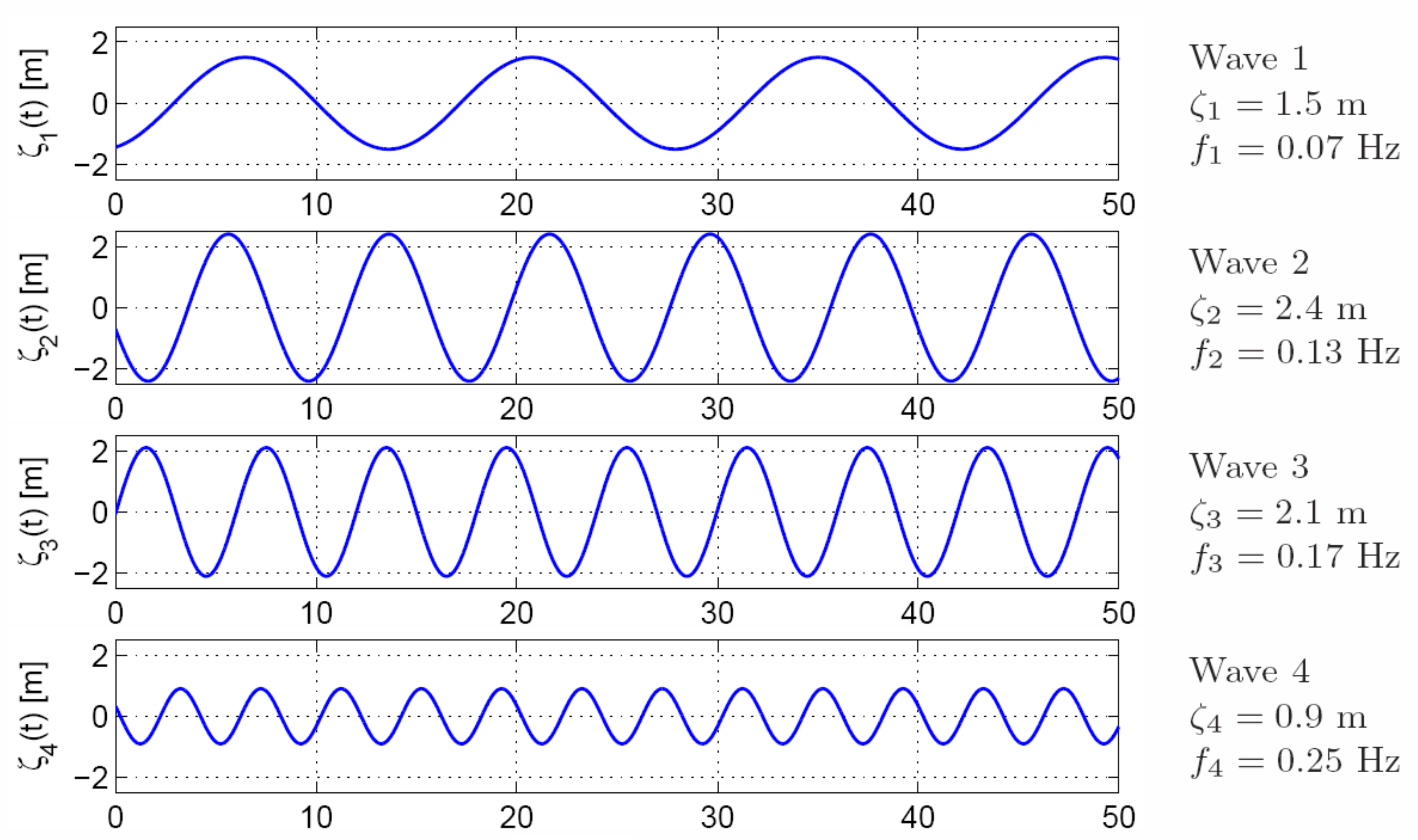
- Maximum wave height: 5.5 m
- Mean wave height: $\frac{1}{15}\sum_{i=1}^{15} H_i$ = 2.9 m
- Significant wave height: $\frac{1}{5}\sum_{i=1}^{5} H_i$ = 4.4 m

Homework: Repeat the example for the wave *period*.

Definition: Average of the largest one thirds of the wave heights

The "transition" to irregular waves

- How do we go from regular waves to irregular waves? Superposition!
  (of regular waves at a fixed position x = 0)



$$\zeta_1(t) = \zeta_{a,1}\cos(-2\pi f_1 t + \epsilon_1)$$

$$\zeta_2(t) = \zeta_{a,2}\cos(-2\pi f_2 t + \epsilon_2)$$

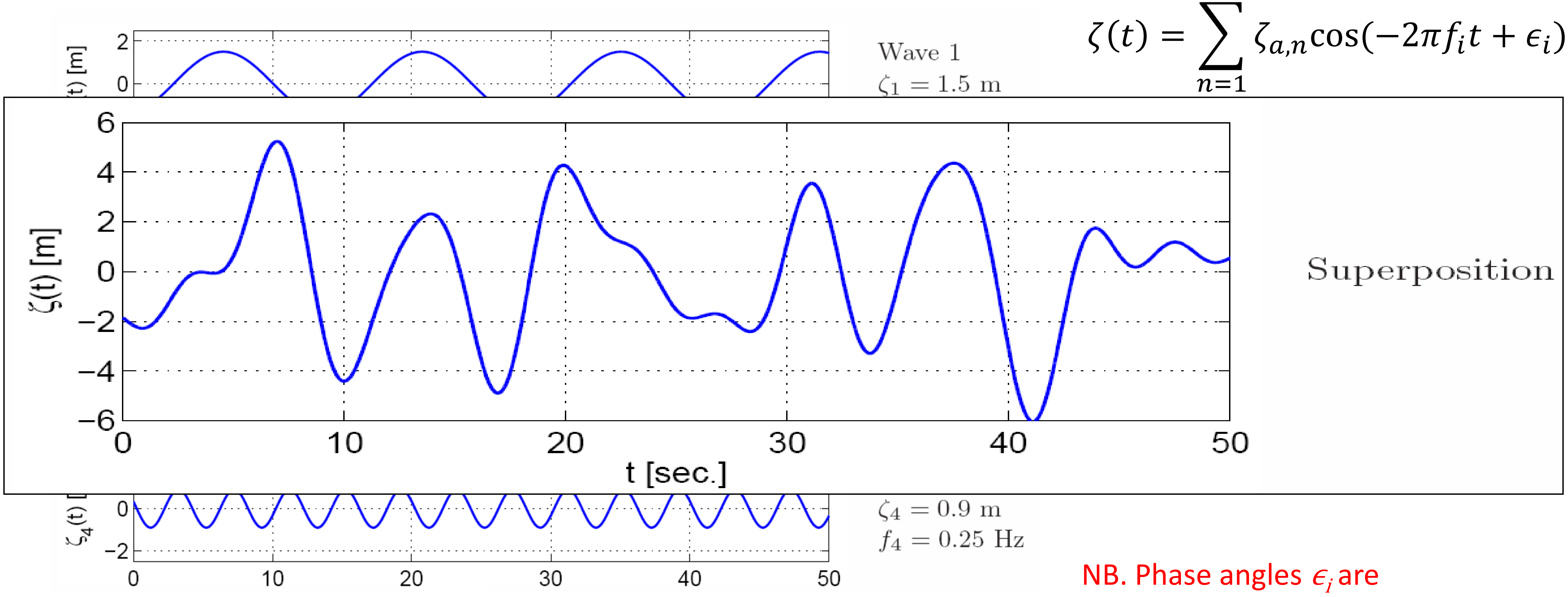
$$\zeta_3(t) = \zeta_{a,3}\cos(-2\pi f_3 t + \epsilon_3)$$

$$\zeta_4(t) = \zeta_{a,4}\cos(-2\pi f_4 t + \epsilon_4)$$

The “transition” to irregular waves

- Superposition of regular waves (at a fixed position x = 0)

$$\zeta(t) = \sum_{n=1}^{4} \zeta_{a,n} \cos(-2\pi f_i t + \epsilon_i)$$



NB. Phase angles ϵ_i are chosen randomly in [0; 2π].

The "transition" to irregular waves

- The mathematical foundation: In Fourier series analysis, any function is expressed over the interval [0-T_H] as the sum:

$$\zeta(t,x) = \sum_{n=1}^{\infty} A_n \cos(k_n x - \omega_n t) + B_n \sin(k_n x - \omega_n t)$$

$$A_n = \frac{2}{T_H} \int_0^{T_H} \zeta(t,x) \cos(k_n x - \omega_n t) dt$$

$$B_n = \frac{2}{T_H} \int_0^{T_H} \zeta(t,x) \sin(k_n x - \omega_n t) dt$$

- Alternatively, we find that

$$\boxed{\zeta(t,x) = \sum_{n=1}^{N} \zeta_{a,n} \cos(k_n x - \omega_n t + \epsilon_n)}$$

(using the trigonometric sum identity)

$$\zeta_{a,n} = \sqrt{A_n^2 + B_n^2}$$

$$\tan \epsilon_n = -\frac{B_n}{A_n}$$

Example in Matlab

A given wave is found as a superposition of nine regular waves. The nine waves are characterised by the following components:

Wave no.	1	2	3	4	5	6	7	8	9
Wave height [m]	5.0	4.3	3.8	3.6	3.3	2.8	2.2	1.2	0.5
Wave period [s]	10.3	12.0	9.4	14.0	7.0	6.2	5.0	4.2	3.3

1. Show the variance diagram of the wave. (NB: Variance of regular wave = $\langle \zeta(t)^2 \rangle = ½\zeta_a^2$, see slide 18)

2. Draw the superposed wave in a period t = [0; 100] s; the phase lags must be *uniformly distributed* between 0 and 2π.

Short-term statistics of irregular waves

Assumptions / properties*

- Stationary conditions (in the stochastic sense) meaning that the probability distribution of the wave elevation does not change;
- Surface elevation, ζ, is Gaussian distributed (= normal distr.) with mean value $\mu = 0$ and standard deviation s; that is, negative and positive values are equally likely,
- Wave crests and troughs - hence, the wave *heights* - are Rayleigh distributed with mean value μ_2 and standard deviation s_2); i.e. *only* positive values.

*In fact, we talk about *assumptions* for real ocean waves, while it will be *properties* for simulated irregular waves because of the Central Limit Theorem…

$$\zeta(t) = \sum_{n=1}^{\boldsymbol{N}} \zeta_{a,n} \cos(-\omega_n t + \epsilon_i)$$

Short-term statistics of irregular waves

Time series (realization)

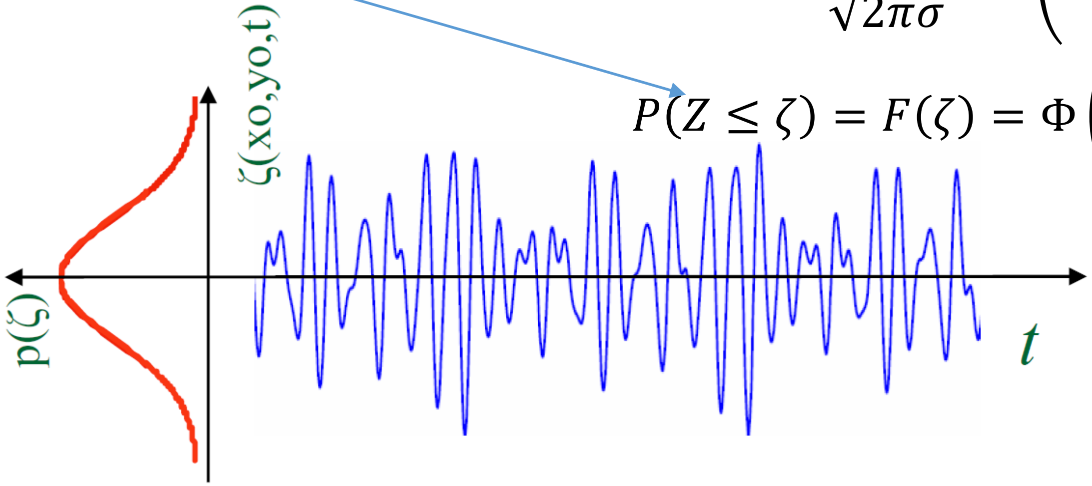
Surface elevation is Gaussian distributed:

Surface elevation at any time t

$$p(\zeta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\left(\frac{\zeta-\mu}{\sigma}\right)^2\right); -\infty < \zeta < \infty$$

$$P(Z \leq \zeta) = F(\zeta) = \Phi\left(\frac{\zeta-\mu}{\sigma}\right)$$

mean value (= 0)
standard deviation



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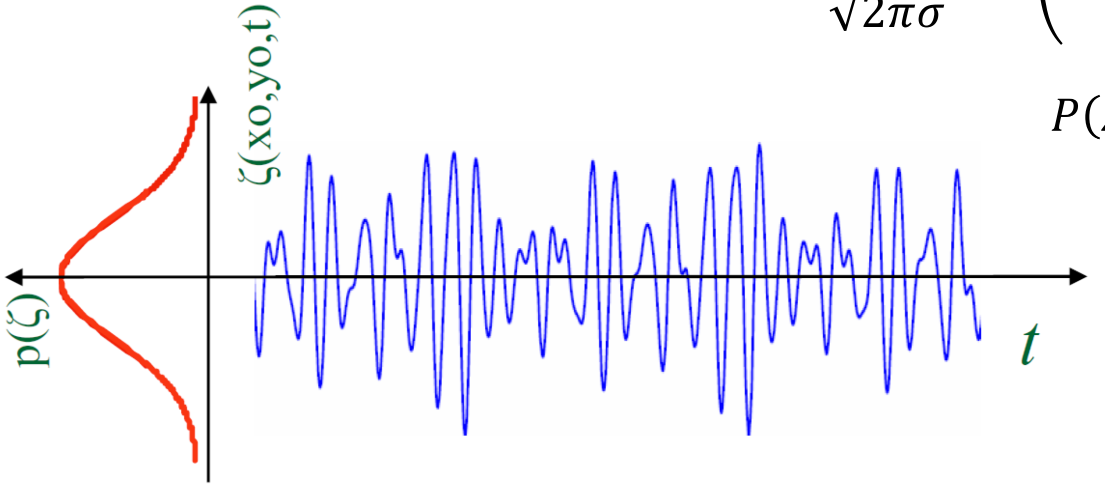
Short-term statistics of irregular waves

Time series (realization)

$$p(\zeta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\left(\frac{\zeta-\mu}{\sigma}\right)^2\right); -\infty < \zeta < \infty$$

$$P(Z \leq \zeta) = F(\zeta) = \Phi\left(\frac{\zeta-\mu}{\sigma}\right)$$

Surface elevation is Gaussian distributed:

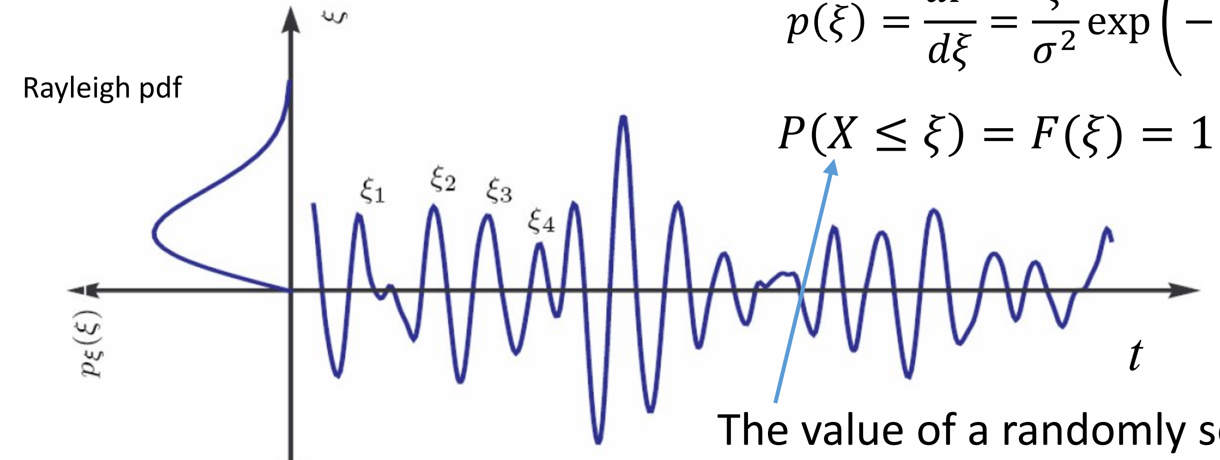


$$p(\xi) = \frac{dF}{d\xi} = \frac{\xi}{\sigma^2} \exp\left(-\frac{1}{2}\frac{\xi^2}{\sigma^2}\right), \qquad \xi > 0$$

$$P(X \leq \xi) = F(\xi) = 1 - \exp\left(-\frac{1}{2}\frac{\xi^2}{\sigma^2}\right), \qquad \xi > 0$$

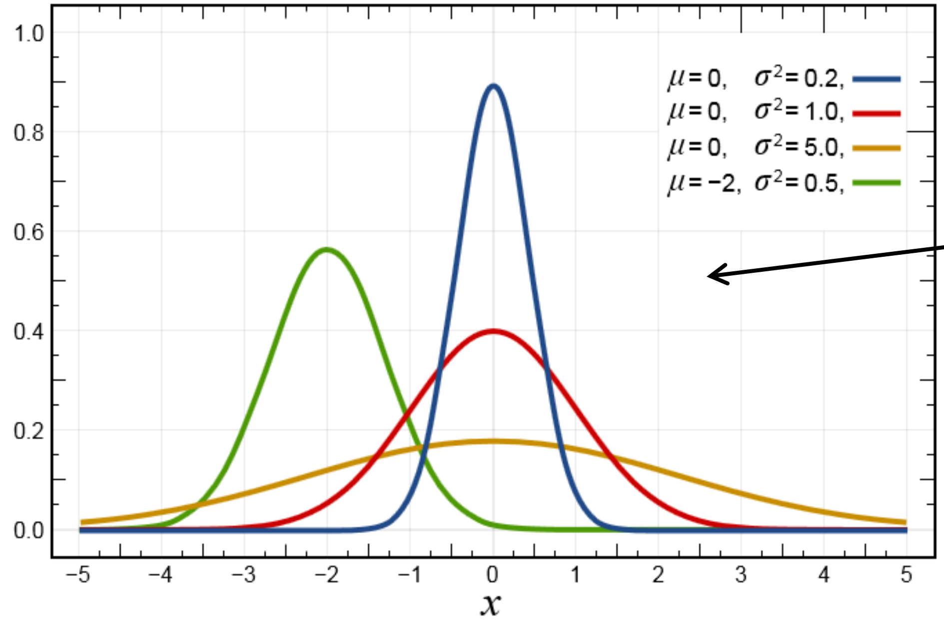
Wave crests (and troughs) follow a Rayleigh distribution:

NB: Requires a narrow-banded process.



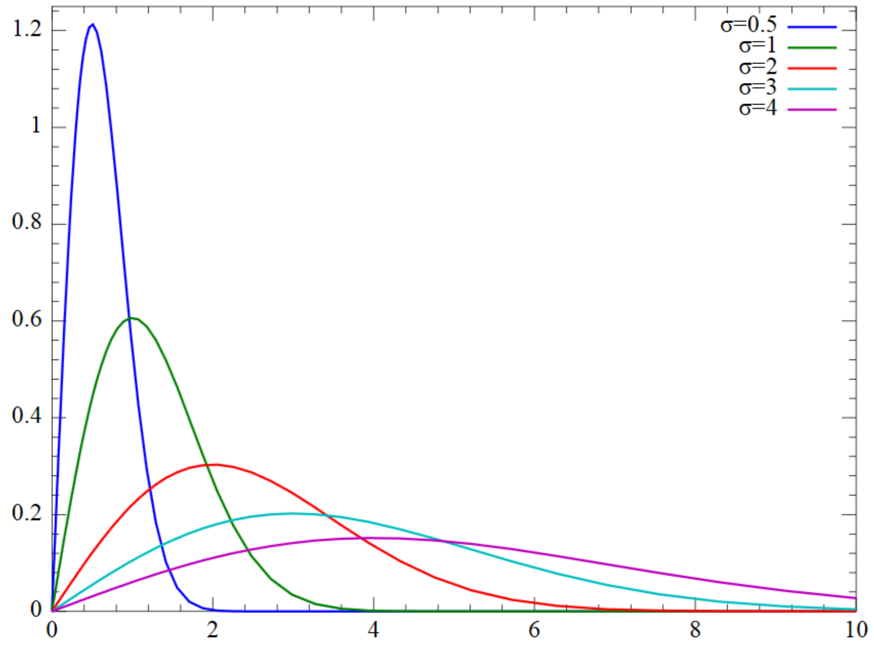
The value of a randomly selected crest

Statistics of irregular waves



Normal distributions (probability density functions)

The realization of surface elevation ("waves") is normally assumed to have zero-mean, and we note that positive and negative values are equally likely



Rayleigh distributions (pdf)

Distribution of wave heights (requires a narrow-banded process); only positive values occur.

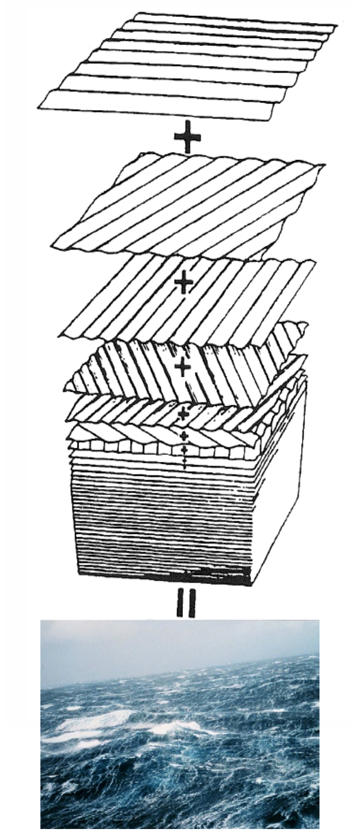
Example: Probability of exceedance

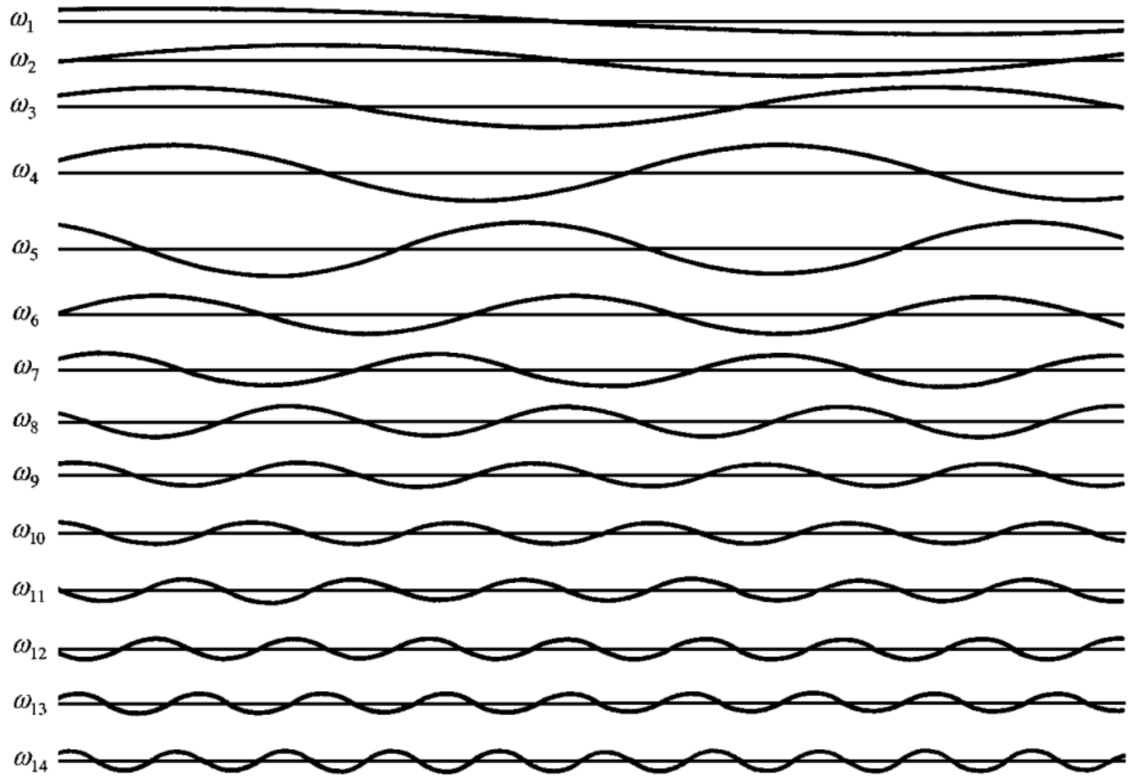
Consider a wave system represented by sea state code 6. For the time history recorded (a total of 23.5 minutes), the variance of the surface elevation was 1.56 m^2, and the significant wave height was 5.0 m.

1. What is the probability that a wave *crest* exceeds the level h_l = 1.5 m?
2. Find the probability that the wave *height* exceeds the significant wave height.
3. ~~Find the probability that the wave *height* exceeds 8.0 m.~~ (see example-pdf)

The concept of a wave spectrum

Recall the energy of the n-*th* wave component $\frac{1}{2}\rho g \zeta_{0,n}^2$





The concept of a wave spectrum

From the energy of the n-*th* wave component ($\frac{1}{2}\rho g \zeta_{0,n}^2$) we can define the wave *energy spectral density*

$$\rho g S_\zeta(\omega_n)\delta\omega = \frac{\rho g \zeta_{0,n}^2}{2}$$

$$S_\zeta(\omega_n) = \frac{\zeta_{0,n}^2}{2\delta\omega}$$

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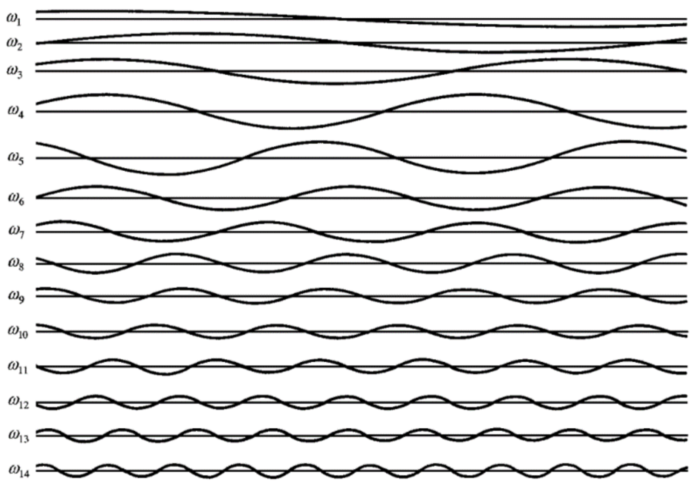
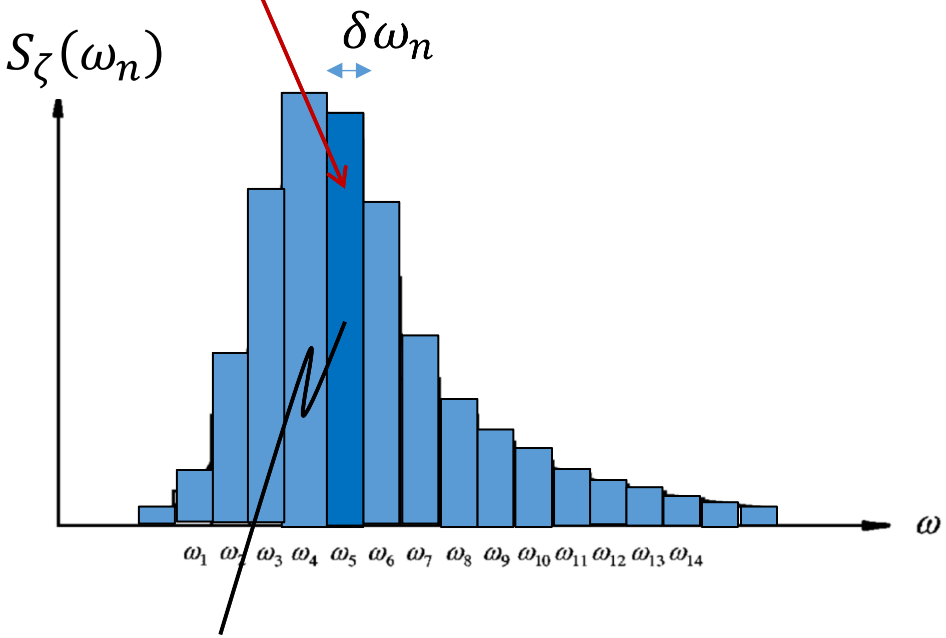




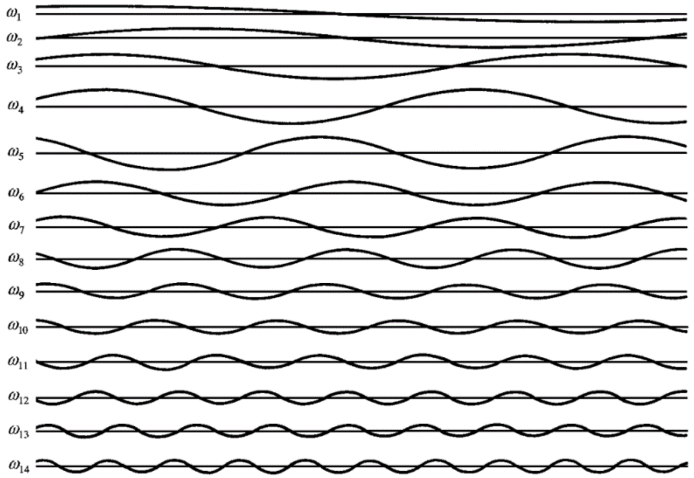
energy density in one single (regular) wave with frequency ω_5

The concept of a wave spectrum

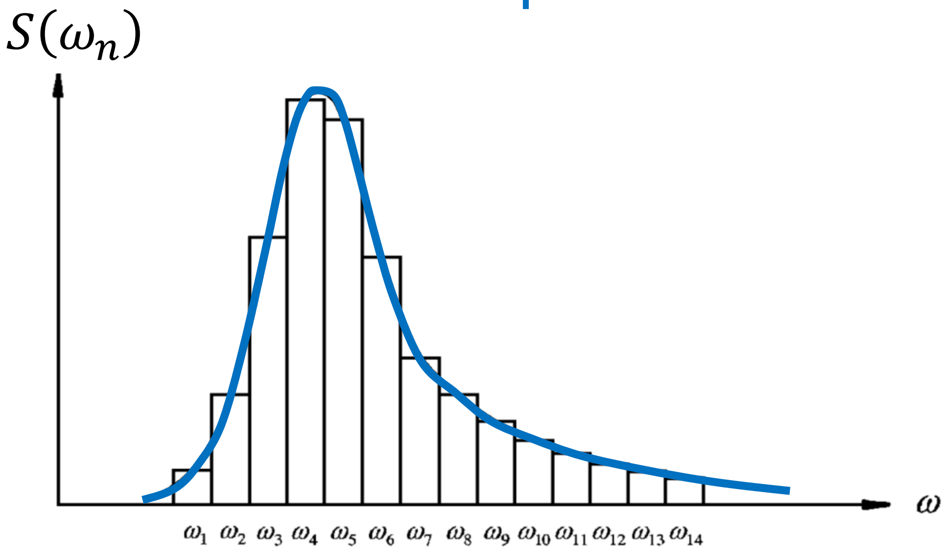
From the energy of the n-*th* wave component ($\frac{1}{2}\rho g \zeta_{0,n}^2$) we can define the wave energy spectral density

$$\rho g S_\zeta(\omega_n)\delta\omega = \frac{\rho g \zeta_{0,n}^2}{2}$$

$$\Rightarrow \quad S_\zeta(\omega_n) = \frac{\zeta_{0,n}^2}{2\delta\omega}$$



The ***wave energy density spectrum***, or simply just the wave spectrum



Note the resemblance to the variance spectrum introduced earlier.

The concept of a wave spectrum

Characteristics and properties of a wave spectrum

- Any given wave spectrum $S(\omega)$ is the complete representation of a certain wave system;

1. The wave system is composed by the (infinite) sum of regular waves, each with component amplitude, by definition, $\zeta_{0,n} = \sqrt{2S(\omega_n)\delta\omega_n}$

2. The total energy[1] (equivalently variance) σ^2 of the wave system is equivalent to the area under the spectrum:

$$\sigma^2 = \sum_n \frac{1}{2}\zeta_{0,n}^2 = \sum_n S(\omega_n)\delta\omega_n = \int_0^\infty S(\omega)d\omega$$

[1]Normalised by ρg

The concept of a wave spectrum

Note that any wave system (responsible for the elevation of a sea surface being a Gaussian process and with wave heights following the Rayleigh distribution) **<u>is fully defined by the wave spectrum</u>**.

This follows from:

- The individual amplitudes (component waves) are given, $\zeta_{0,n} = \sqrt{2S(\omega_n)\delta\omega_n}$
- The mean value is known (zero)
- The standard deviation (i.e., variance) is known, $\sigma^2 = \int_0^\infty S(\omega)d\omega$
- Additional *characterizing parameters* can be derived…

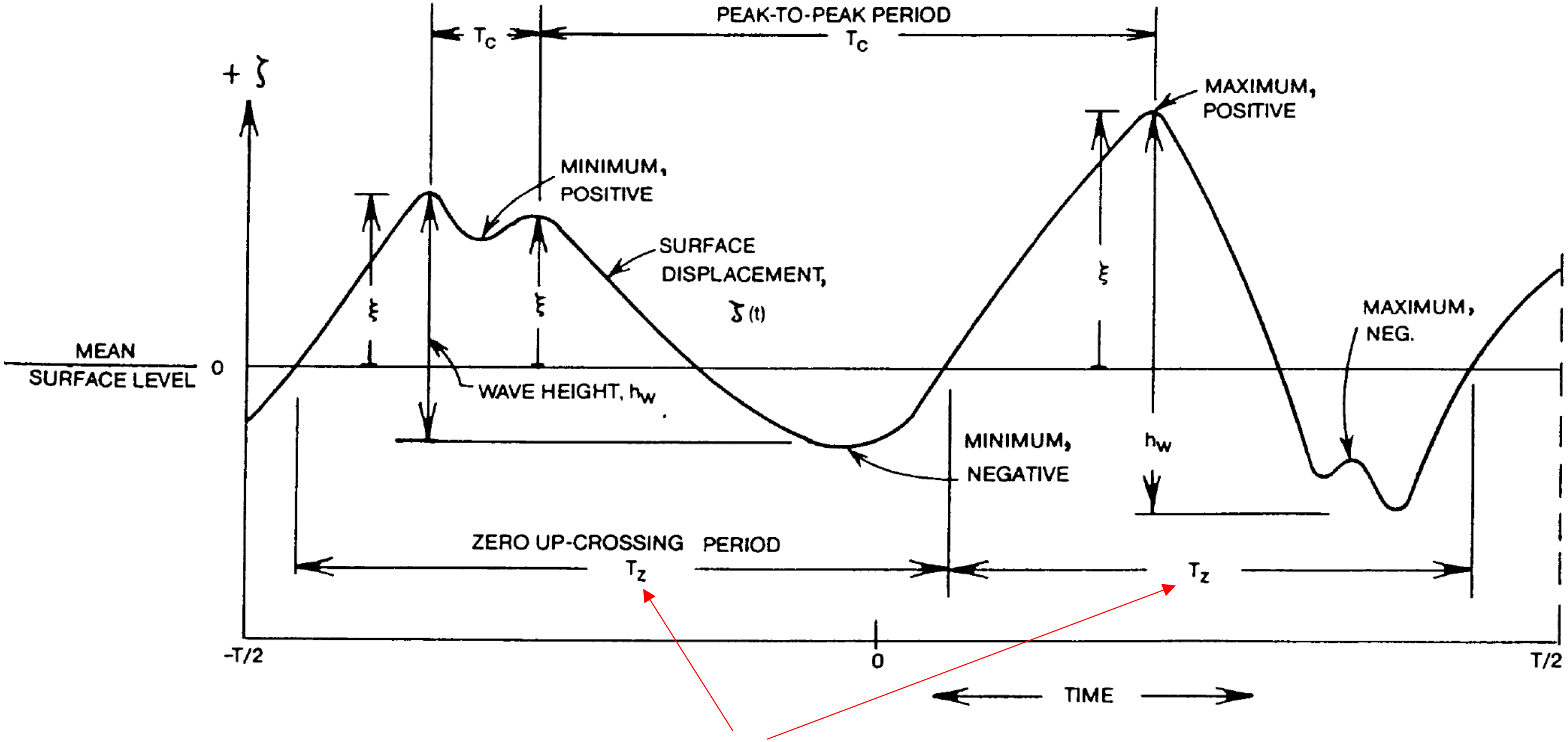
Spectral moments

- A number of *integral wave parameters* [next slides] can be expressed through the n-th order spectral moment, defined by:

$$\boxed{m_n = \int_0^\infty \omega^n \, S_\zeta(\omega) \, d\omega}$$

- Note that the variance of the surface elevation becomes identical to m_0, equivalently the area under the spectrum.
- The integral wave parameters correspond to characteristic wave parameters that are statistically representative for the wave system in study…

Illustration of characteristic periods



Clearly, two succeeding "identical" periods from a wave record are not identical in numerical values…

WARNING: The parameters may be referred to by different symbols (letters)!

Integral (average) wave parameters

- Significant wave height: $H_s = H_{1/3} = 4\sqrt{m_0}$
- Average period of component waves: $T_E = T_{-1} = 2\pi \frac{m_{-1}}{m_0}$
  ("mean energy period")
- Period corresponding to average frequency: $T_1 = 2\pi \frac{m_0}{m_1}$
  ("mean wave period", sometimes denoted T_m)
- Average period between zero-upcrossings: $T_z = T_2 = 2\pi \sqrt{\frac{m_0}{m_2}}$
- Average period between maxima (crests): $T_c = 2\pi \sqrt{\frac{m_2}{m_4}}$
- Modal period ("peak period", sometimes denoted by T_0 or T_m): $T_p = \frac{2\pi}{\omega_p}, \omega_p = \max\big(\mathrm{S}(\omega)\big)|_\omega$

Ocean wave statistics

Why do we need *measurements* (statistics and/or predictions) of ocean waves?

- Design purposes (ships, wind power, aqua farms, coastal structures…)
- Routing of ships
  - before (avoidance of storms)
  - during (operation guidance)
  - after (performance evaluations)
- Environmental studies (oceanography, climate)

How can we obtain information on local scale?, on global scale?

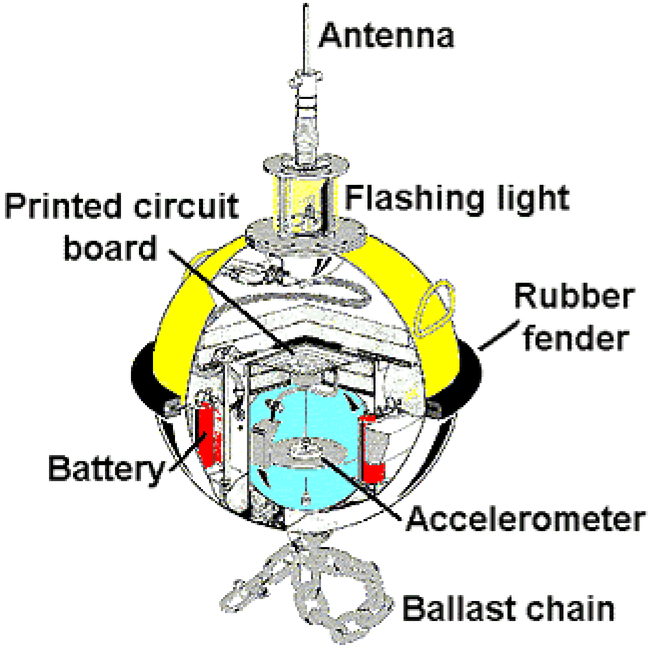
- Measurements at fixed position (but we need to measure for a long time), and observations from ships
- Numerical models (assimilated with measured data)

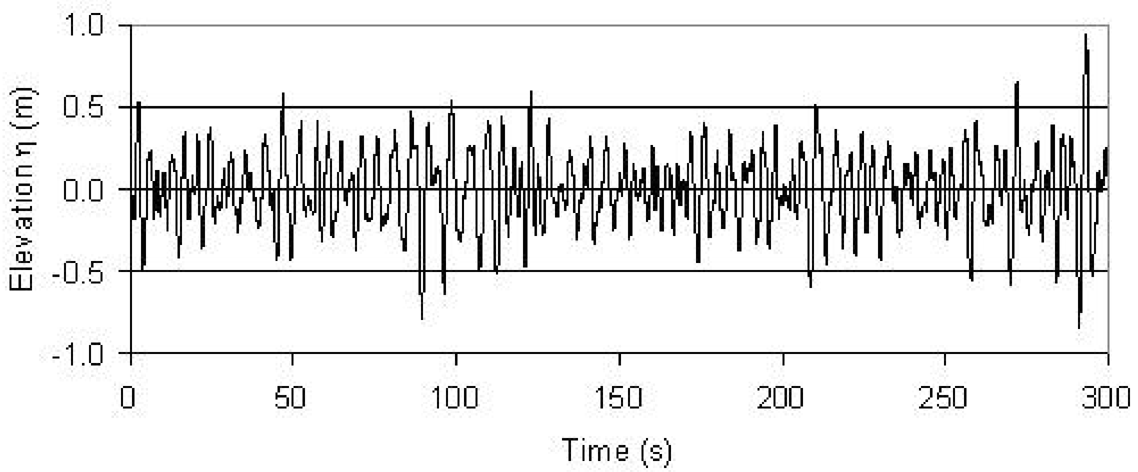
The interaction between waves and a structure will be the topics of lectures 10 and 11…

Ocean wave statistics

Wave measurements and data

- The most reliable measurements are obtained with wave buoys
- Other means include, e.g., satellite, wave radar, ship-wave-buoy analogy, and visual observation

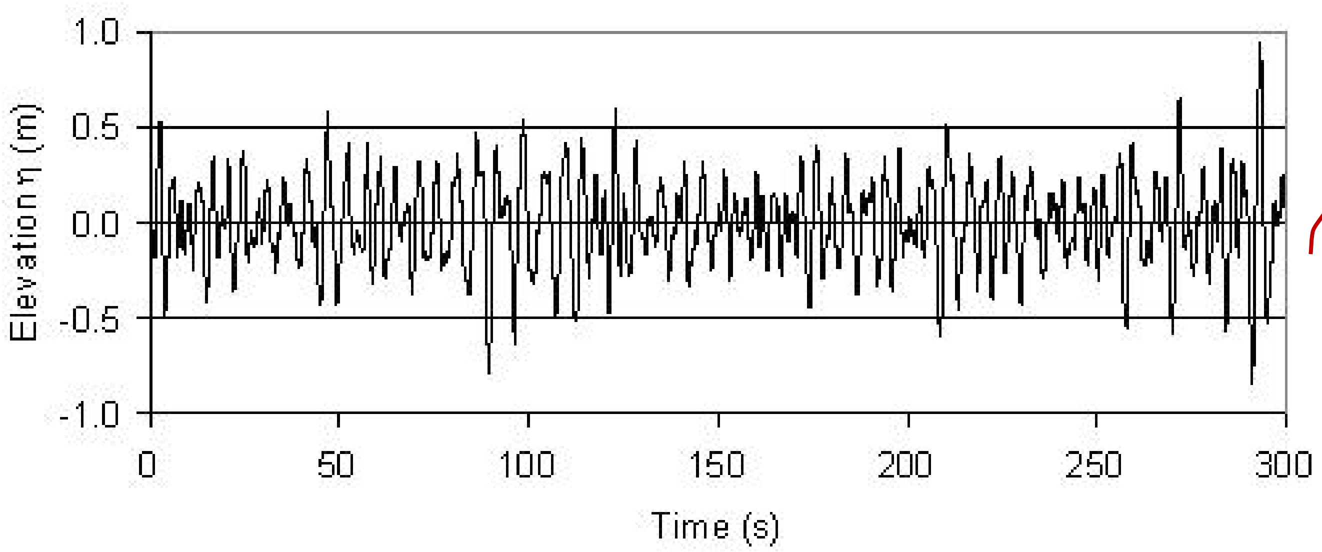


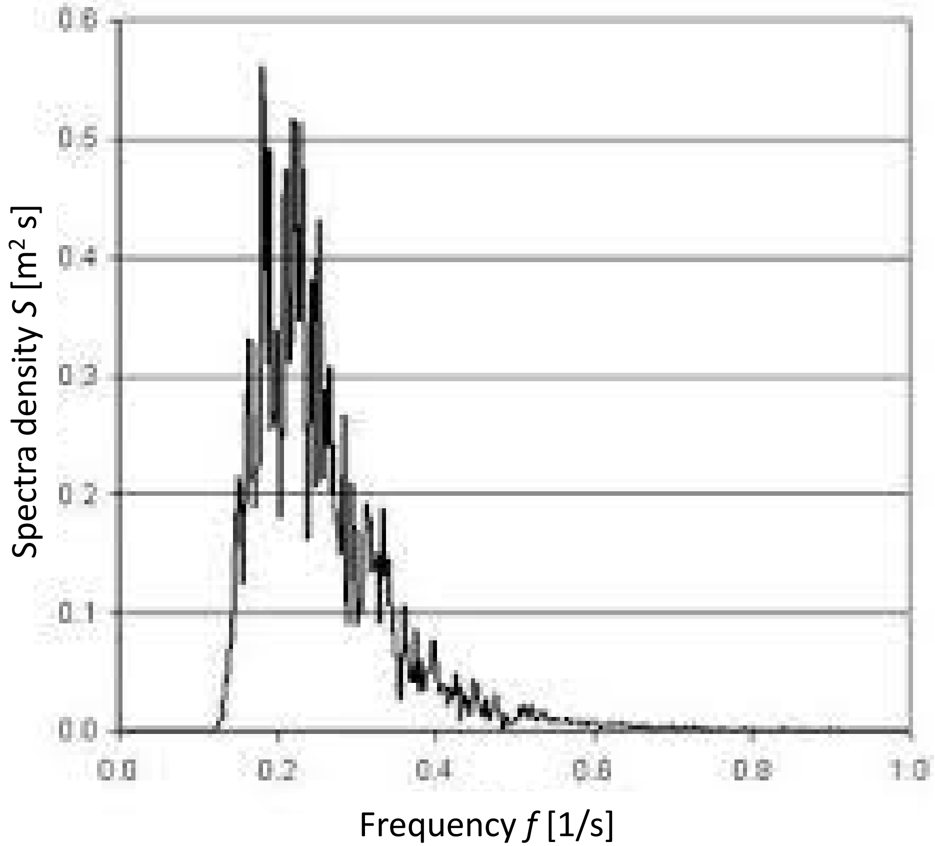


Ocean wave statistics

Measured point wave spectra

- A *measured* spectrum can be estimated from **FFT (Fast Fourier transformation) techniques**[1]





[1]Built-in tools exist in Matlab, Python, … but this is not the present focus. (See, e.g., 41226 Advanced Wave Hydrodynamics)

Hindcast (prediction) of ocean waves over larger areas

Spectral wave models and assimilation (computing)

- Third generation spectral wave models can produce the directional wave spectra, typically computed in grid points covering *larger ocean areas*
- The ERA5 dataset is the result of a spectral wave model where observations / measurements have been assimilated, see https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-single-levels?tab=overview
- The ERA5 database is freely available for academic and educational uses

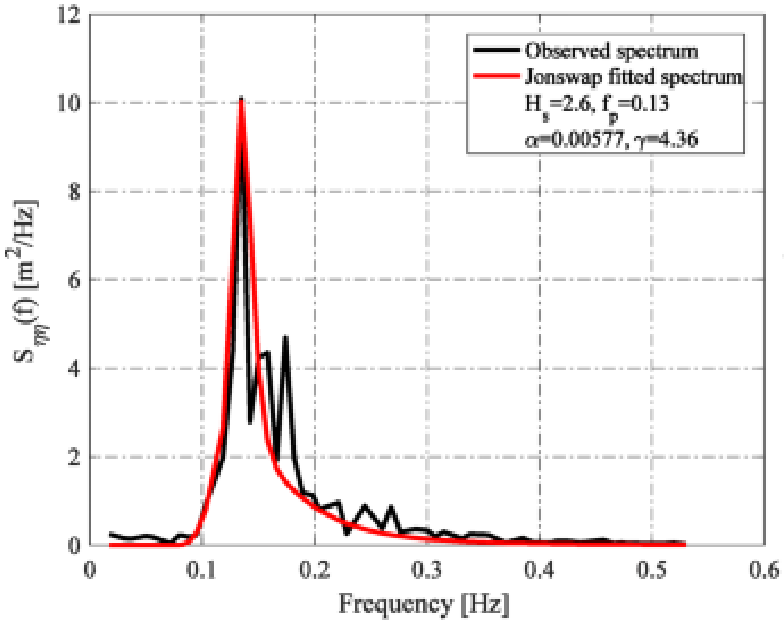
Idealised (= parameterised) wave spectra

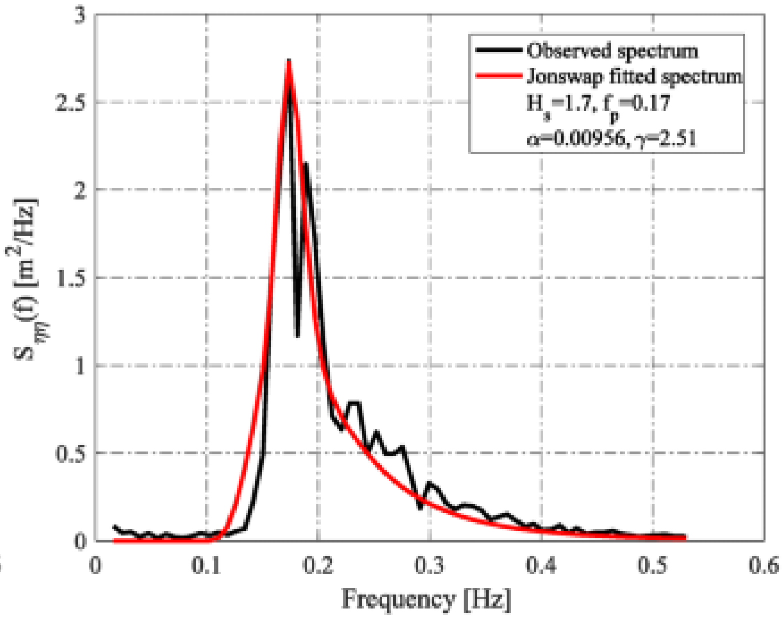
- We have seen that statistics of *wave parameters* are available.
- In our further analyses (L10 + L11), it will be evident that the analysis of wave-structure interactions, <u>requires</u> the actual **wave spectrum** to be available!

- However, it is not very practical to store statistics of and process huge numbers of wave spectra (storing capacity! two parameters versus complete directional spectrum)
- Instead, idealised wave spectra are introduced…

Idealised (= parameterised) wave spectra

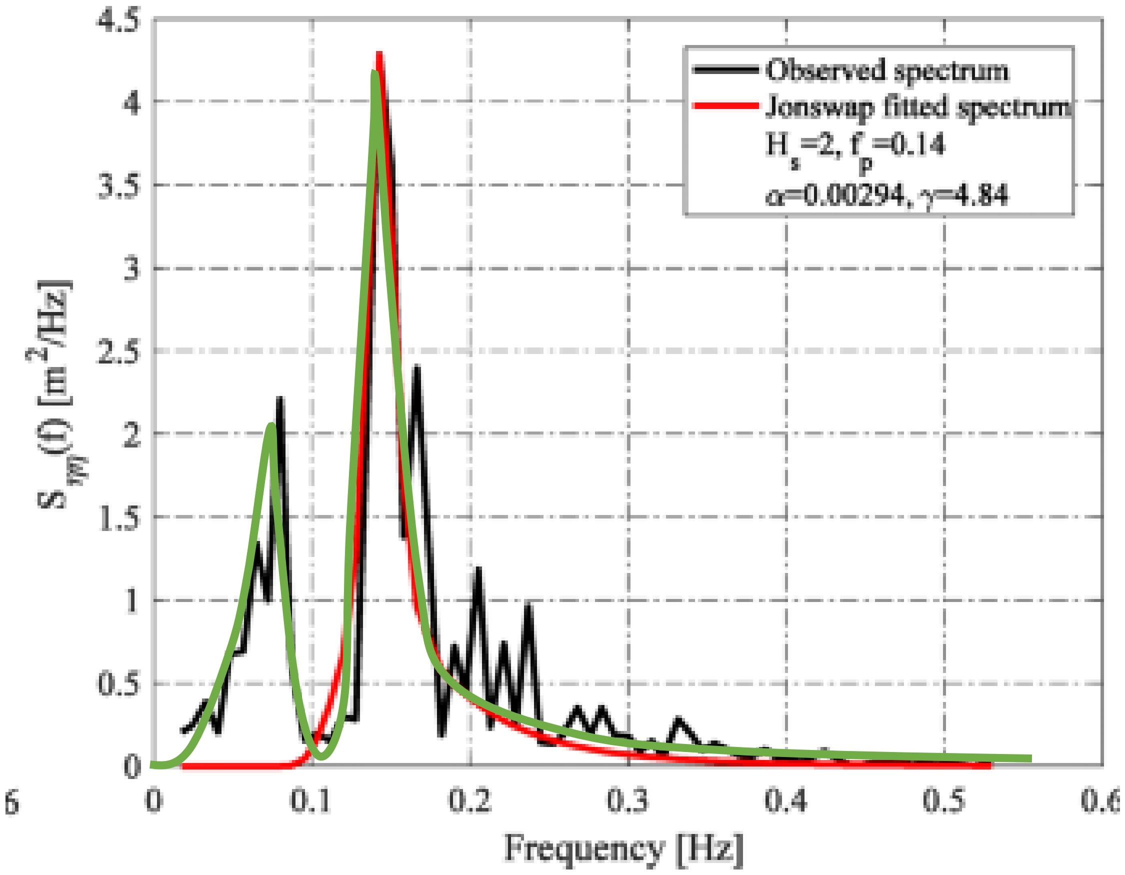
- Based on the characteristic (statistical) wave parameters of a given ocean area, a typical wave system of the particular area can be described by a *parameterised wave spectrum* taking as input the statistical wave parameters
- Well-known parameterised wave spectra are the Bretschneider spectrum, the JONSWAP spectrum (including Pierson-Moskowitz spectrum), and the Ochi 6-parameter spectrum





Idealised versus observed wave spectra

Why do we need (several…) different idealised wave spectra?



Fitting with the Ochi 6-parameter spectrum…

Parameterised wave spectra

- Bretschneider spectrum:

unit of the spectral density

$$S(\omega) = \frac{A}{\omega^5} \exp \frac{-B}{\omega^4} \quad \text{[m}^2 \text{ s/rad]}$$

$$A = \frac{H_s^2}{4\pi} \left(\frac{2\pi}{T_z} \right)^4$$

$$B = \frac{1}{\pi} \left(\frac{2\pi}{T_z} \right)^4$$

- Pierson-Moskowitz spectrum (requires info about wind speed!):

$$S(\omega) = \frac{8.1 \times 10^{-3} g^2}{\omega^5} \exp\left[-0.74 \left(\frac{g}{V\omega} \right)^4 \right]$$

- V is the wind speed [m/s] as measured 19.5 m above the surface
- g is the acceleration of gravity [m/s^2]

Parameterised wave spectra

- JONSWAP spectrum (requires info about wind speed and fetch):

$$S_j(\omega) = \frac{\alpha g^2}{\omega^5} \exp\left[-\frac{5}{4}\left(\frac{\omega_p}{\omega}\right)^4\right]\gamma^r$$

$$r = \exp\left[-\frac{(\omega-\omega_p)^2}{2\sigma^2\omega_p^2}\right]$$

$$\alpha = 0.076\left(\frac{U_{10}^2}{F\,g}\right)^{0.22}$$

$$\omega_p = 22\left(\frac{g^2}{U_{10}F}\right)^{1/3}$$

$$\gamma = 3.3$$

$$\sigma = \begin{cases} 0.07 & \omega \leq \omega_p \\ 0.09 & \omega > \omega_p \end{cases}$$

F is the fetch

(see more on: https://wikiwaves.org/Ocean-Wave_Spectra)

Parameterised wave spectra

- JONSWAP spectrum (requires info about wind speed and fetch):

$$S_j(\omega) = \frac{\alpha g^2}{\omega^5} \exp\left[-\frac{5}{4}\left(\frac{\omega_p}{\omega}\right)^4\right]\gamma^r$$

$$r = \exp\left[-\frac{(\omega-\omega_p)^2}{2\sigma^2\omega_p^2}\right]$$

$$\alpha = 0.076\left(\frac{U_{10}^2}{F\,g}\right)^{0.22}$$

$$\omega_p = 22\left(\frac{g^2}{U_{10}F}\right)^{1/3}$$

$$\gamma = 3.3$$

$$\sigma = \begin{cases} 0.07 & \omega \le \omega_p \\ 0.09 & \omega > \omega_p \end{cases}$$

F is the fetch

(see more on: https://wikiwaves.org/Ocean-Wave_Spectra)

- 6-parameter Ochi spectrum:

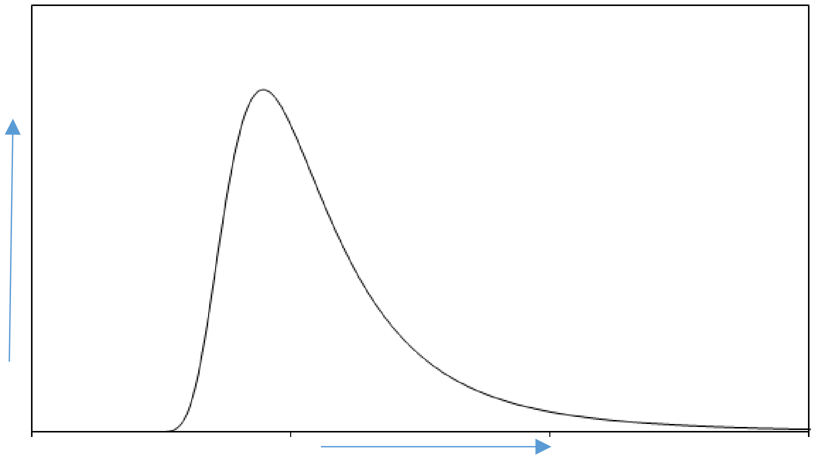
$$S(\omega) = {}^1/_4 \sum_k \frac{\left(\frac{4\lambda_k+1}{4}\omega_{p,k}^4\right)^{\lambda_k}}{\Gamma(\lambda_k)} \frac{H_{s,k}^2}{\omega^{4\lambda_k+1}} \exp\left[-\frac{(4\lambda_k+1)}{4}\left(\frac{\omega_{p,k}}{\omega}\right)^4\right]$$

k = 1,2

Parameterised wave spectra

Validation of numerical implementation

- Compute the spectral moments (m_0, m_1,…) and check that the derived wave parameters match the input, cf. slide 46
- Make sure that correct frequency is used if required; angular freq. versus temporal freq. (rad/s vs. Hz)



Angular frequency [rad/s] *or* temporal frequency [Hz]

$$S(\omega)d\omega = S(f)df$$

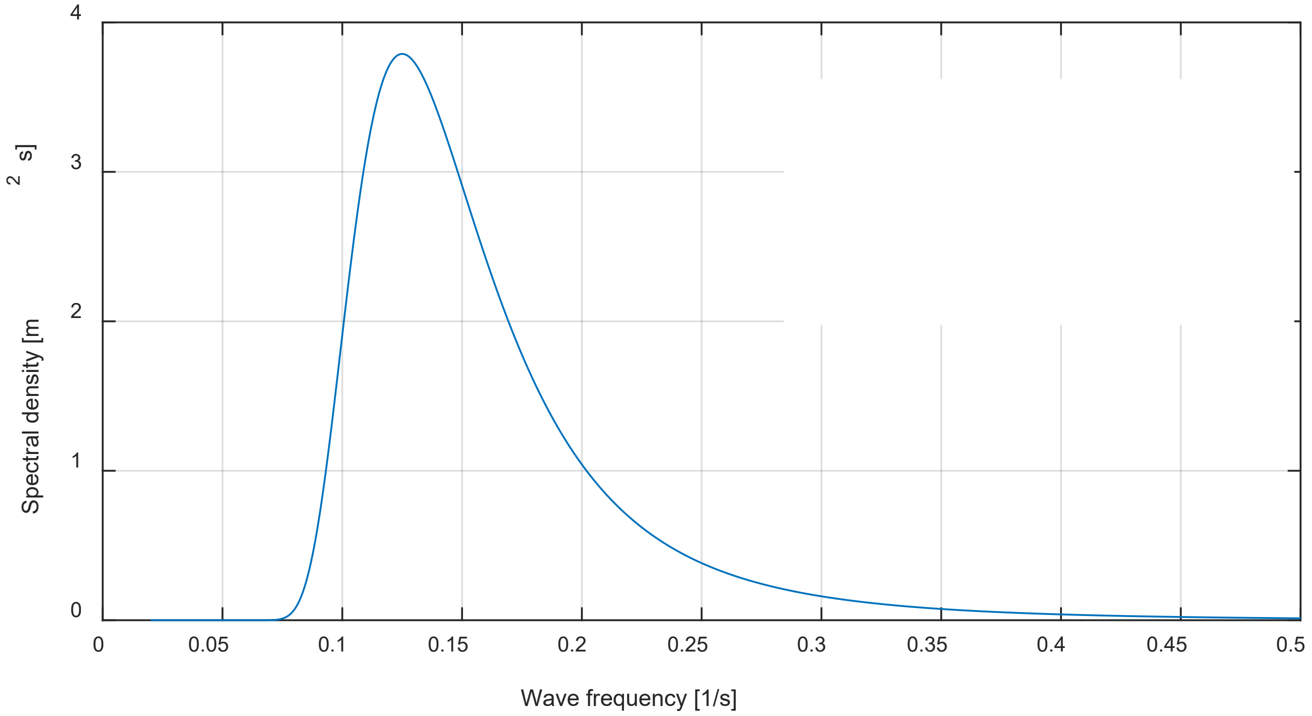
$$\omega = 2\pi f;\ d\omega/df = 2\pi$$

$$\mathbf{S(f) = S(\omega)\ 2\pi}$$

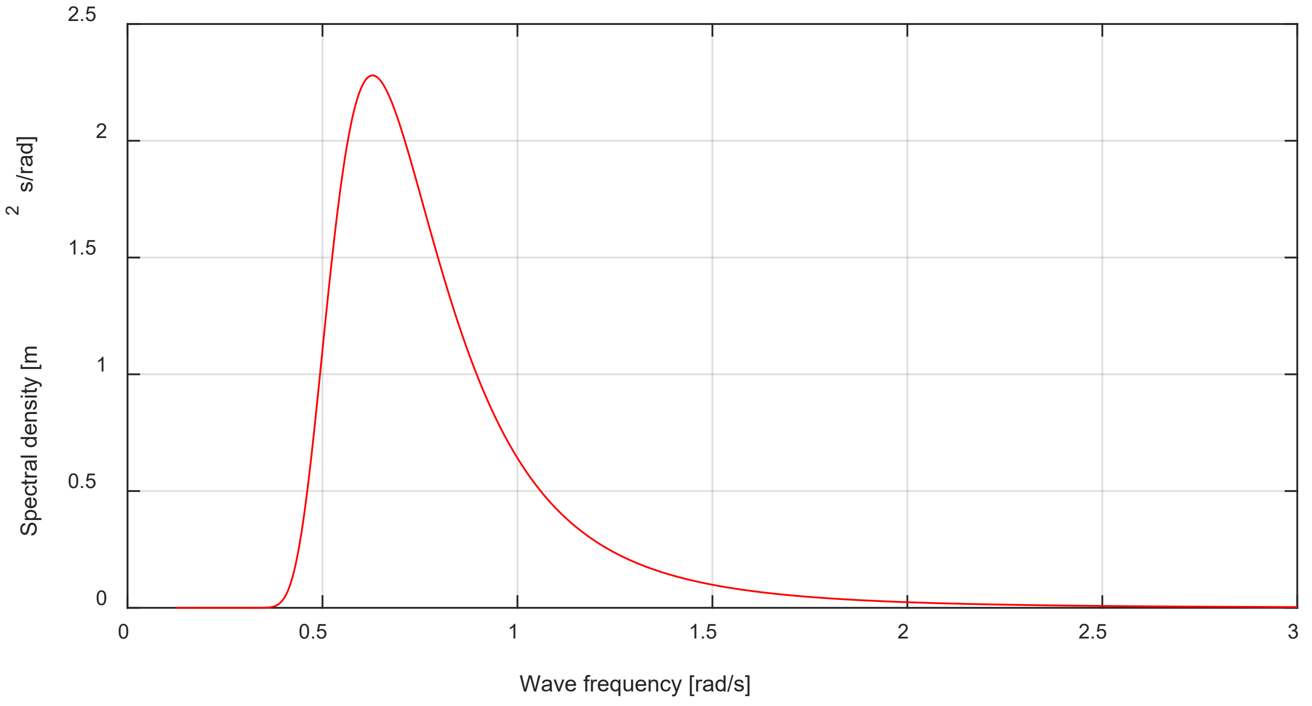
Example in Matlab

Implementation of Ochi-Hubble spectrum (one-peaked version) with H_s = 2.3 m and T_p = 8 s

Matlab…



Example: Visual/crude estimation….



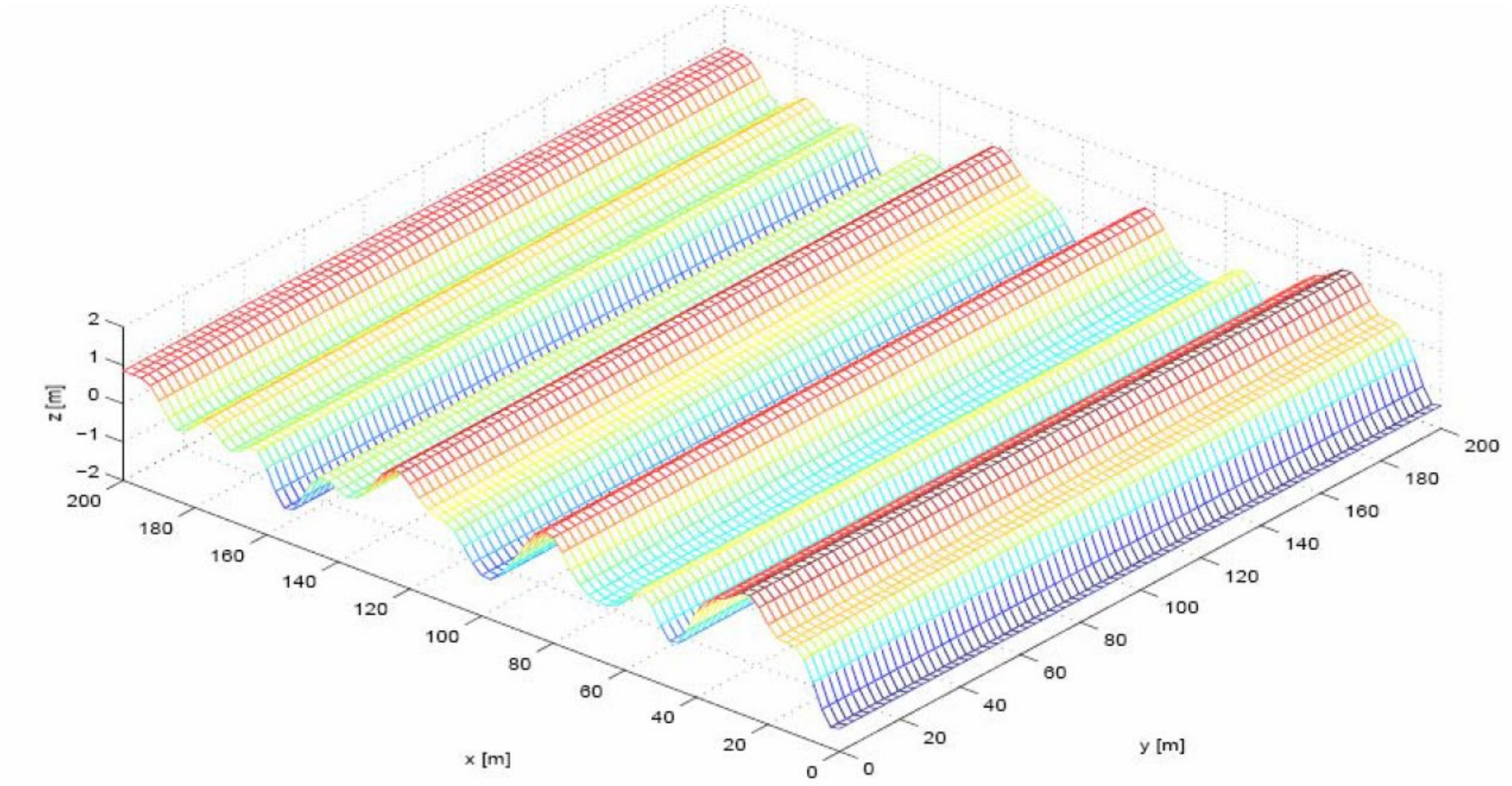
- Determine H_s
- How many exceedances of the level h_l = 2.0 m can be expected in a period of one hour?

True parameters:

< result >

Irregular (long-crested) waves

In many situations, we restrict ourselves to long-crested waves



Assignment B

Assessment of a semisubmersible

Sea state and wave-induced responses

- All questions must be answered, and the report must be handed in no later than **5th May (by midnight)**
- Work in groups of max 5 people (self-enroll on Learn)
- All remaining "post-lecture" submodules used
- Assignment (text) becomes available on DTU Learn

Appendix

Ocean wave statistics – Marsden areas

marsden.sq

T_z

H_s

```
34    ; ATLANTIC OCEAN SCATTER SQ. 34

  2.0   4.5   5.5   6.5   7.5   8.5   9.5  10.5  11.5  12.5 ; Tz
  0.5     1    12    35    36    18     5     1     0     0     0  ;  Hs, Nobs(Tz)
  1.5     0     6    51   114   109    59    21     6     1     0  ;  Hs, Nobs(Tz)
  2.5     0     1    16    60    90    72    37    14     4     1  ;  Hs, Nobs(Tz)
  3.5     0     0     4    19    38    40    26    12     4     1  ;  Hs, Nobs(Tz)
  4.5     0     0     1     5    12    15    12     7     3     1  ;  Hs, Nobs(Tz)
  5.5     0     0     0     1     3     5     4     3     1     1  ;  Hs, Nobs(Tz)
  6.5     0     0     0     0     1     1     1     1     1     0  ;  Hs, Nobs(Tz)
  7.5     0     0     0     0     0     1     1     0     0     0  ;  Hs, Nobs(Tz)
```

Example: Derived probabilities…

- $P(3 < H_s < 4 \text{ and } 8 < T_z < 9\,) = \frac{38}{1000} = 0.038 = 3.8\%$
- $P(H_s > 5\,) = \frac{1+4+7+6+4+2+1}{1000} = 0.025 = 2.5\%$

But how can *statistics* be used for design and operation? (NB: Statistics will always contain a given maximum/maxima; not really what we want, so then what to do?!)

Ocean wave statistics – Marsden areas

Fitting a parameterised PDF to the *empirical distribution* of Hs…

- PDF of H_s is derived from the total observations in each H_s bin
- Fitting of the empirical PDF can be made by, e.g., Matlab, using *all* observations of H_s

T_z →

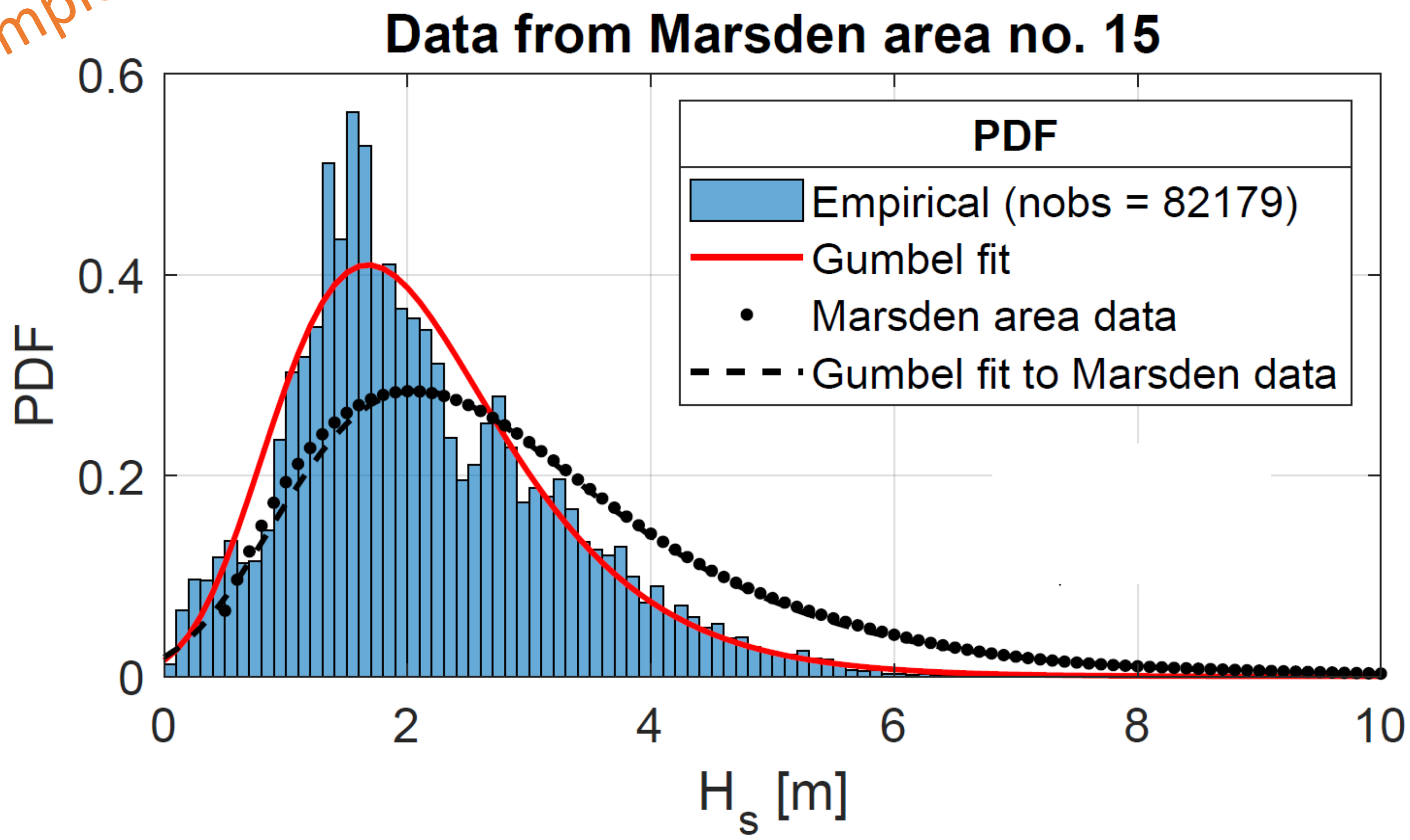
H_s ↓

2.0	4.5	5.5	6.5	7.5	8.5	9.5	10.5	11.5	12.5	; Tz		
0.5	1	12	35	36	18	5	1	0	0	0	; Hs, Nobs(Tz)	sum(Nobs) = 108
1.5	0	6	51	114	109	59	21	6	1	0	; Hs, Nobs(Tz)	sum(Nobs) = 367
2.5	0	1	16	60	90	72	37	14	4	1	; Hs, Nobs(Tz)	… etc.
3.5	0	0	4	19	38	40	26	12	4	1	; Hs, Nobs(Tz)	… etc.
4.5	0	0	1	5	12	15	12	7	3	1	; Hs, Nobs(Tz)	
5.5	0	0	0	1	3	5	4	3	1	1	; Hs, Nobs(Tz)	
6.5	0	0	0	0	1	1	1	1	1	0	; Hs, Nobs(Tz)	
7.5	0	0	0	0	0	1	1	0	0	0	; Hs, Nobs(Tz)	

HsAll = [0.5*ones(1,108) , 1.5*ones(1,367) , 2.5*ones(…) … 7.5 *ones(1,2)]

Ocean wave statistics – Marsden areas

Example



From Nielsen and Ikonomakis (2021): Wave conditions encountered by ships—A report from a larger shipping company based on ERA5, *Ocean Engineering*, **237:** 109584.